

OSCAR RODRIGUES

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PROFESSIONAL EXPERIENCE

ADAS Operations Engineer – Team Lead, Stellantis | Windsor, ON Mar 2022 - Present

- Detected and proactively responded to road construction projects through **web scraping** using **Python**, resulting in a **19%** competitive advantage and enhanced customer safety.
- Quantified and addressed erratic steering in autonomous vehicles due to trajectory discrepancy, demonstrating expertise in **exploratory data analysis (EDA)** and **logistic regression** model development.
- Led execution of internal process development and documentation for a team of six engineers, resulting in a **40%** reduction in new hire onboarding time.

Junior Data Analyst, SPM Automation | Oldcastle, ON Mar 2021 - Mar 2022

- Designed and deployed an **ETL pipeline** that used **Python** scripting to automate data extraction and established network configuration to upload data to an **S3 bucket**.
- Developed a data warehouse solution that allowed integration between **Tableau** and **Amazon S3** via **Athena**, enabling real-time monitoring and visualization of machine data through dashboards.
- Leveraged sensor data to provide clients with proactive maintenance recommendations and identify unauthorized operator adjustments, saving the company approximately **\$1.3 million** in warranty claims.

PROJECTS

Quantifying Vehicle Steering Discrepancies Jun 2023 - Aug 2023

- Restructured and normalized complex JSON data effectively using **Pandas**, enabling structured analysis and tabulation of key signals.
- Employed **Matplotlib** to visualize steering angles and vehicle trajectories, facilitating the visual identification of steering discrepancies.
- Utilized **feature engineering** to optimize a **logistic regression** model designed for predicting steering discrepancies, successfully achieving an accuracy rate of **93.4%**.

Proactive Road Construction Detection System Oct 2022 - Dec 2022

- Leveraged **BeautifulSoup** to web scrape construction-related data from Department of Transportation (DOT) websites across states, ensuring real-time access to critical information.
- Applied **data wrangling** techniques to process and clean the acquired data, ensuring its accuracy and consistency for further analysis.
- Implemented a **MySQL** database system to store and manage construction data, enabling precise identification and disabling of potentially hazardous areas available to the autonomous driving system.

EDUCATION & CERTIFICATIONS

Electrical Engineering, Honors, BAsC, University of Windsor | Windsor, ON Aug 2020

IBM Data Science Specialization, Coursera | Online Learning Platform Sept 2023

Supervised Machine Learning: Regression and Classification, Coursera | Online Learning Platform Dec 2023

TECHNICAL SKILLS

- Programming Languages: Python (NumPy, Pandas, ScikitLearn) | SQL | MATLAB
- Machine Learning: Regression | Classification | Clustering
- Data Visualization: Tableau | Microsoft PowerBI | Plotly (Dash) | Databricks | Looker